for any choice of $i \neq j$. We do not consider any restriction on the connectedness of the wells, in particular some U_i and U_j may be rank-one connected. By polar decomposition, we can suppose that $U_1, \dots, U_l$ are symmetric and positive definite. Let $K = \cup_{i=1}^{l} \mathrm{SO}(3)U_i$. We denote by $W\colon \mathbb{R}^{3\times 3} \to [0, +\infty]$ the elastic energy density. We assume W to be Borel measurable and to satisfy the following hypotheses:

(W1) $W(F) = 0 \iff F \in K$,

(W2) W is C^2 in a neighborhood of K,

(W3) W is frame indifferent, i.e., $W(RF) = W(F)$ for every $R \in \mathrm{SO}(3)$ and for every $F \in \mathbb{R}^{3\times 3}$,

(W4) $W(F) \geq C f_q(\mathrm{dist}(F, K))$ for every $F \in \mathbb{R}^{3\times 3}$,

where $C > 0$ and $f_q(t) = t^2 \wedge t^q$ and $q \in [0, 2]$. Note that far away from the set K the energy density may have linear or sublinear growth. We will denote by Q_j the quadratic form $D^2W(U_j)$.

For $h > 0$ we consider a thin plate $\Omega_h = S \times (-\frac{h}{2}, \frac{h}{2}) = S \times hI$, where $S \subset \mathbb{R}^2$ is an open, connected and bounded set with Lipschitz boundary and $I = (-\frac{1}{2}, \frac{1}{2})$. For some results we will need higher regularity of the boundary of S. More precisely, we will request that

$$\begin{gathered}\text{there is a closed subset } \Sigma \subset \partial S \text{ with } \mathcal{H}^1(\Sigma) = 0 \text{ such that} \\ \text{the outer unit normal } \vec{n} \text{ to } S \text{ exists and is continuous on } \partial S \backslash \Sigma\,.\end{gathered} \tag{2}$$

This property is called *condition* $(*)$ in [17]. We denote by Ω the rescaled plate, that is $\Omega = \Omega_1$.

Given $\alpha \in [2, +\infty)$, set $\gamma = \frac{\alpha}{2}$. We choose $p > 1$ and $\eta\colon (0, +\infty) \to (0, +\infty)$ such that for some constant $C > 0$:

(P1) $\eta(h) \geq C h^{\frac{\alpha}{3}}$ for every $h > 0$,

(P2) $\eta(h) h^{\gamma(1-\frac{2}{p})-1} \to 0$ as $h \to 0$,

(P3) $p > \frac{6}{5}$, if $q < 2$.

Conditions (P1) and (P3) ensure that the penalty term is strong enough to provide suitable compactness estimates (see Proposition 3.1) whereas condition (P2) guarantees that the penalty term is negligible at the limit. Note that (P1)–(P3) are compatible, since for every $\alpha \geq 2$ we have $1 - \gamma(1 - \frac{p}{2}) < \frac{\alpha}{3}$ for p large enough.

The symbol $\nabla_h y$ denotes the rescaled gradient of y, while $\nabla_h^2 y$ is the rescaled Hessian of y. They are defined as follows:

$$(\nabla_h y)_{ij} = \begin{cases} \partial_j y_i & \text{if } j \neq 3\,, \\ \frac{1}{h}\partial_3 y_i & \text{otherwise}\,, \end{cases}$$

$$(\nabla_h^2 y)_{ijk} = \begin{cases} \partial_{jk} y_i & \text{if } j, k \neq 3\,, \\ \frac{1}{h}\partial_{3k} y_i & \text{if } j = 3\,, k \neq 3\,, \\ \frac{1}{h}\partial_{j3} y_i & \text{if } j \neq 3\,, k = 3\,, \\ \frac{1}{h^2}\partial_{33} y_i & \text{if } j = 3\,, k = 3\,. \end{cases}$$

We set

$$\mathcal{H}_p(\Omega; \mathbb{R}^3) = \{y \in W^{1,2}(\Omega; \mathbb{R}^3)\colon \nabla_h^2 y \in L^p(\Omega; \mathbb{R}^{3\times 3\times 3})\}\,,$$

and for $\alpha \geq 2$ we define the α-rescaled energy $E_h^\alpha \colon W^{1,2}(\Omega, \mathbb{R}^3) \to \mathbb{R}$ as follows:

$$E_h^\alpha(y) = \begin{cases} \dfrac{1}{h^\alpha} \displaystyle\int_\Omega W(\nabla_h y)\, dx + \dfrac{\eta^p(h)}{h^\alpha} \displaystyle\int_\Omega |\nabla_h^2 y|^p \, dx & \text{if } y \in \mathcal{H}_p(\Omega; \mathbb{R}^3)\,, \\ +\infty & \text{otherwise}\,. \end{cases}$$

We explain now some useful matrix notation that we will use throughout the paper. Given a general matrix $A \in \mathbb{R}^{3\times 3}$ we will write A' to denote the 2×2 submatrix obtained removing the third row and column. Similarly, we will often write x' in place of (x_1, x_2) and ∇' instead of ∇_{x_1,x_2}. We will use the super(sub)scripts to denote submatrices of A in the following way: every missing subscript index is a removed row while every missing superscript index is a removed column. For example, $A^{1,2}$ is the 3×2 submatrix given by the first two columns of A while $A_{1,2}$ is the 2×3 submatrix given by the first two rows of A. Whenever we will sum or multiply matrices and vectors with different dimension we will imply that the smaller one is naturally embedded in the bigger space by adding zeros in the missing entries. For example, if $F \in \mathbb{R}^{2\times 2}$ and $G \in \mathbb{R}^{3\times 3}$ the expression $F + G$ means $\iota(F) + G$ where

$$\iota : \mathbb{R}^{2\times 2} \hookrightarrow \mathbb{R}^{3\times 3}\,, \quad F \mapsto \begin{pmatrix} F & 0 \\ 0 & 0 \end{pmatrix}\,.$$

We will denote by I_h the matrix

$$I_h = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & h \end{pmatrix}\,.$$

1.2 Main results

The main objective of this paper is to extend the rigorous derivation of a hierarchy of plate models obtained by Friesecke, James, and Müller in [16, 15] for a single well to the case of a multiwell elastic energy. In order to do so, we need to introduce the limiting models centered at a well K_j. We define

$$\bar{Q}_j \colon \mathbb{R}^{2\times 2} \to \mathbb{R} \quad F \mapsto \min_{a \in \mathbb{R}^3} Q_j(U_j^{-1}(\mathrm{sym}(F) + a \otimes e_3 + e_3 \otimes a))\,.$$

This is a quadratic form that generalizes the one defined in [15, Equation (8)], taking into account that linearization may arise around a possibly nontrivial deformation gradient U_j. The minimum in $\bar{Q}_j$ exists thanks to some coercivity property of Q_j that will be proved later (see Lemma 2.2).

1.2.1 The Kirchhoff regime $\alpha = 2$

For $\alpha = 2$, we will prove the Γ-convergence of E_h^α to the Kirchhoff functional

$$E_j^{\mathrm{K}}(y) = \frac{1}{24} \int_S \bar{Q}_j(\nabla y^T \nabla \nu)\, dx\,, \tag{3}$$

where $y \in W^{2,2}(S; \mathbb{R}^3)$ satisfies the constraint $(\nabla y)^T \nabla y = (U_j^2)'$ and ν is the unique vector such that $(\nabla y, \nu) U_j^{-1} \in \mathrm{SO}(3)$ a.e. in S (whose existence is guaranteed by Lemma 3.6). Note that y is an isometric immersion for the flat metric $(U_j^2)'$. The Γ-convergence result is stated in the following theorem.

Theorem 1.1 (Γ-convergence ($\alpha = 2$))**.** *Suppose that S satisfies (2).*

(i) For any sequence $(y_h) \subset W^{1,2}(\Omega;\mathbb{R}^3)$ that satisfies $E_h^2(y_h) \le C$ for every $h > 0$ there exist $y \in W^{2,2}(S;\mathbb{R}^3)$ and $j \in \{1,\dots,l\}$ such that

(a) $\nabla y^T \nabla y = (U_j^2)'$,

(b) $\nabla_h y_h \to (\nabla y, \nu)$ in $L^2(\Omega;\mathbb{R}^{3\times 3})$ (up to a subsequence), where ν is defined as in Lemma 3.6.

(ii) For any sequence $(y_h) \subset W^{1,2}(\Omega;\mathbb{R}^3)$ as in (i) it holds

$$\liminf_{h\to 0} E_h^2(y_h) \ge E_j^{\mathrm{K}}(y)\,.$$

(iii) For any $y \in W^{2,2}(S;\mathbb{R}^3)$ such that $\nabla y^T \nabla y = (U_j^2)'$ for some index $j \in \{1,\dots,l\}$, there exists a sequence $(y_h) \subset W^{1,2}(\Omega;\mathbb{R}^3)$ such that (b) holds true and

$$\lim_{h\to 0} E_h^2(y_h) = E_j^{\mathrm{K}}(y)\,.$$

1.2.2 The Von Kármán regimes $\alpha > 2$

When $\alpha > 2$, the limiting models are expressed in terms of u and v, the limits of the rescaled in-plane and out-of-plane averaged displacements around a well K_j, called u_h and v_h, respectively. They are defined as follows:

$$u_h\colon S \to \mathbb{R}^2 \qquad x' \mapsto \min\left\{h^{-\gamma}, h^{2-2\gamma}\right\}\begin{pmatrix} w_h(x')\cdot U_j e_1 \\ w_h(x')\cdot U_j e_2\end{pmatrix}\,, \tag{4}$$

$$v_h\colon S \to \mathbb{R} \qquad x' \mapsto h^{1-\gamma} w_h(x')\cdot U_j e_3\,, \tag{5}$$

where we recall that $\gamma = \frac{\alpha}{2}$ and

$$w_h\colon S \to \mathbb{R}^3 \quad x' \mapsto \int_I (y_h - U_j\,\mathrm{Id}_h\, x)\,dx_3\,.$$

Observe that u_h and v_h are the components of the averaged displacement w_h in the basis given by $\{U_j^{-1}e_1, U_j^{-1}e_2, U_j^{-1}e_3\}$. This may not look as a natural choice, since a basis of tangent vectors to the embedded midplane $U_j(S\times\{0\})$ is given by $\{U_j e_1, U_j e_2\}$ and the normal direction is given by $U_j^{-1}e_3$. However, since $\{U_j^{-1}e_i : i = 1,2,3\}$ is the dual basis of $\{U_j e_i : i = 1,2,3\}$, this alternative simplifies both the statement and the computations, and gives a completely equivalent result (see Remark 3.10). When $2 < \alpha < 4$, we retrieve the Γ-convergence to the constrained Von Kármán functional, namely,

$$E_j^{\mathrm{CVK}}(v) = \frac{1}{24}\int_S \bar{Q}_j(\nabla^2 v)\,dx\,,$$

whenever $v \in W^{2,2}(S)$ is such that there exists $u \in W^{1,2}(S;\mathbb{R}^2)$ that solves

$$\nabla u^T + \nabla u + |U_j^{-1}e_3|^2 \nabla v \otimes \nabla v = 0\,. \tag{6}$$

As in the single well case, this constraint means that u and v satisfy a matching isometry condition up to the second order; however, the metric now depends on the well, and it is not necessarily the Euclidean one. More precisely, if one defines for $\varepsilon > 0$

$$y_\varepsilon = U_j\begin{pmatrix} x' \\ 0\end{pmatrix} + \varepsilon U_j^{-1}e_3 v + \varepsilon^2 U_j^{-1}\begin{pmatrix} u \\ 0\end{pmatrix}\,,$$

then (6) is equivalent to $\nabla y_\varepsilon^T \nabla y_\varepsilon = (U_j^2)' + O(\varepsilon^3)$.